



SpaceMaps

Explore the universe through a familiar,
interactive map.

BigRed//Hacks 2026 · Navigation

• Milky Way (illustration)

Cluster

Omega Centauri

Nebula 47 Tucanae

• Eta Carinae

You are here

Crab Nebula



THE PROBLEM

Space distances are numbers nobody can feel.

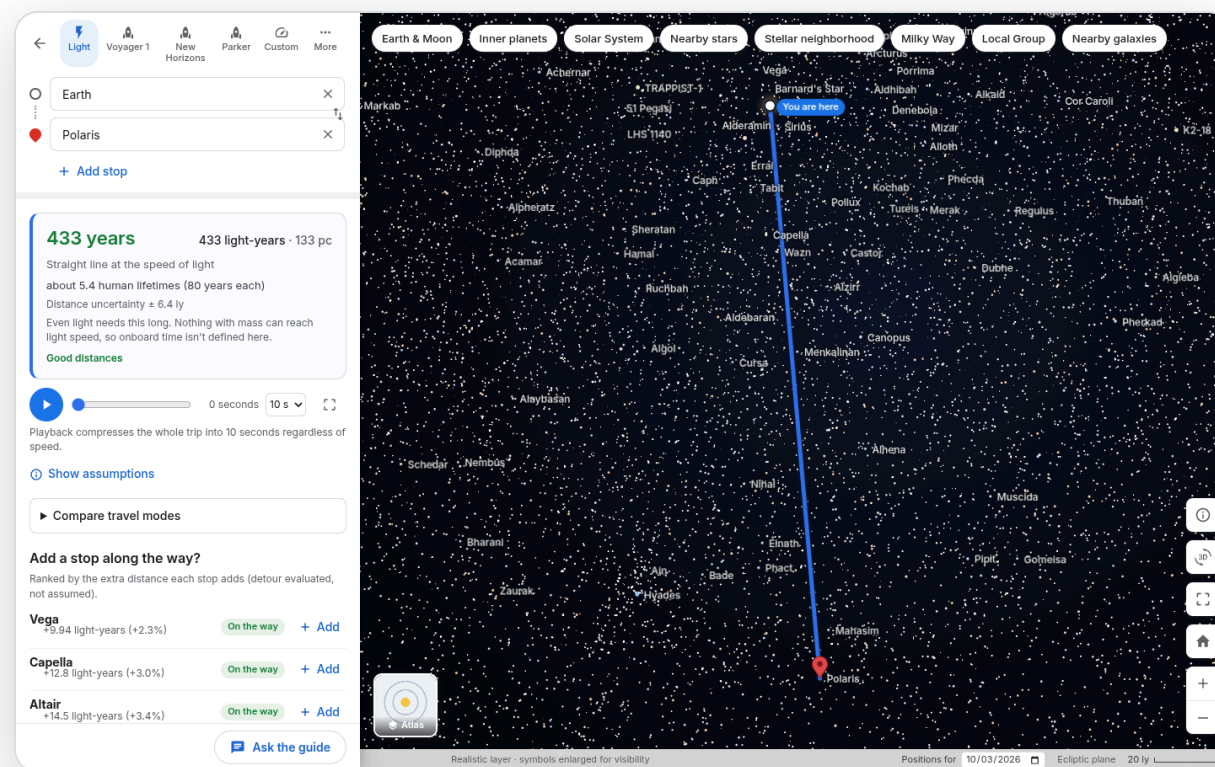
- "Polaris is 433 light-years away." What does that actually mean for a trip?
- Planetarium apps show the sky, not *how far* or *how long*.
- But everyone already knows one spatial interface: **the map**. Search, Directions, compare modes, read the ETA.

What if the universe had directions?

THE PRODUCT

A mapping app for the universe

- One continuous map from the Earth–Moon system to nearby galaxies
- Search 618 objects and 109,000 stars
- Place cards with sourced facts and images
- Directions, travel modes, multi-stop trips, playback
- Realistic and Atlas layers



Earth → Polaris: 433 light-years

433 years

at light speed

7.7 million yr

at Voyager 1's real speed
(JPL, heliocentric)

437 yr / 62 yr

at 0.99 c: Earth time /
onboard (relativity)

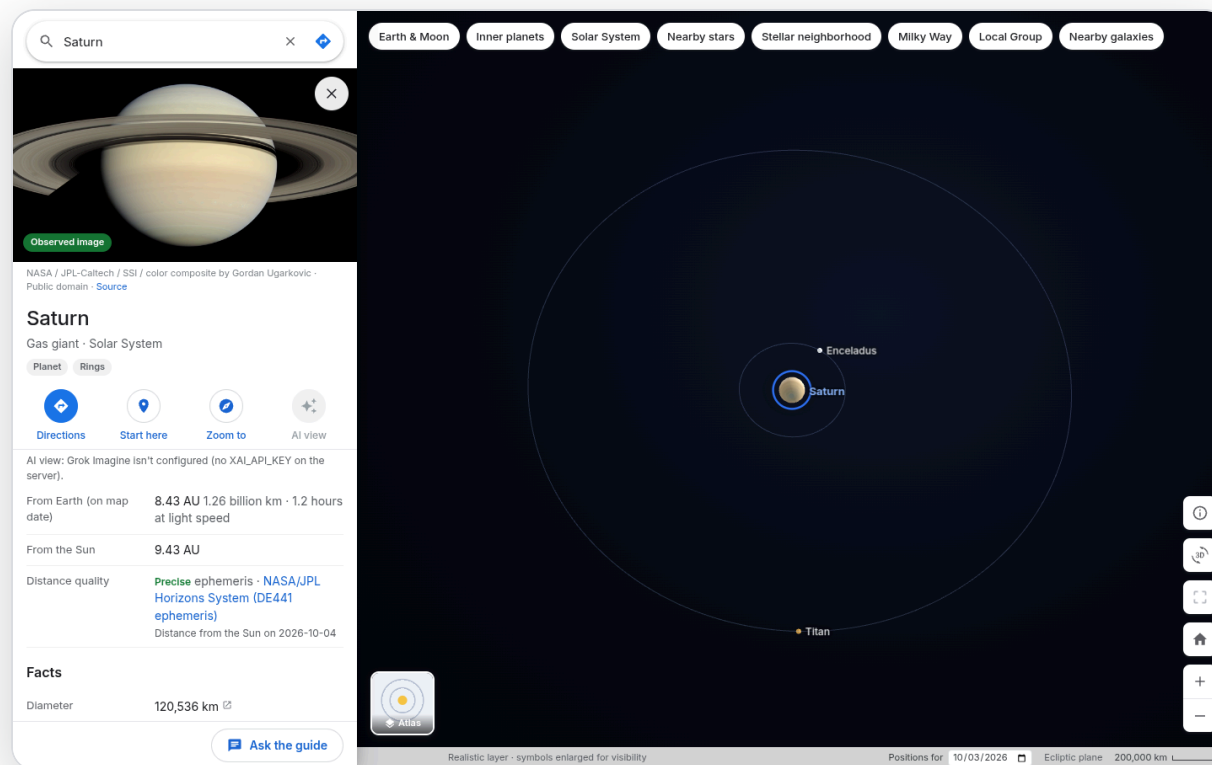
3.4 months

USS Enterprise, warp 9
(fiction)

- True 3D distance between catalog positions, with its uncertainty (± 6.4 ly)
- Constant-speed comparison: no acceleration, gravity or target motion, and the app says so
- "On the way" is computed, never guessed: adding Vega costs **+2.3%**

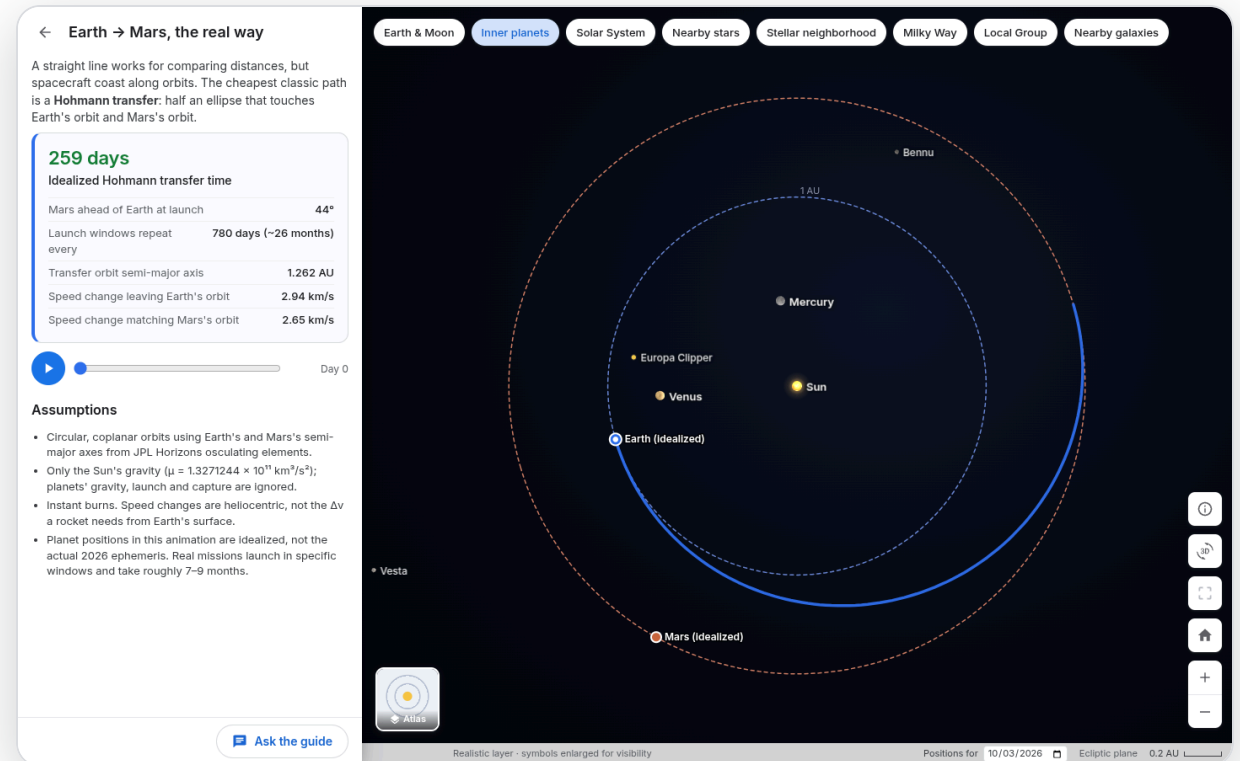
Every number has a source

- NASA/JPL Horizons, SIMBAD, HYG v4.4, NASA Exoplanet Archive, cited papers
- Parallaxes inverted only when the error is $\leq 20\%$; otherwise the object is "route unavailable", with the reason
- No fake routes to cosmological objects (3C 273, TON 618)
- Imagery labeled: Observed image · Scientific illustration · AI reconstruction



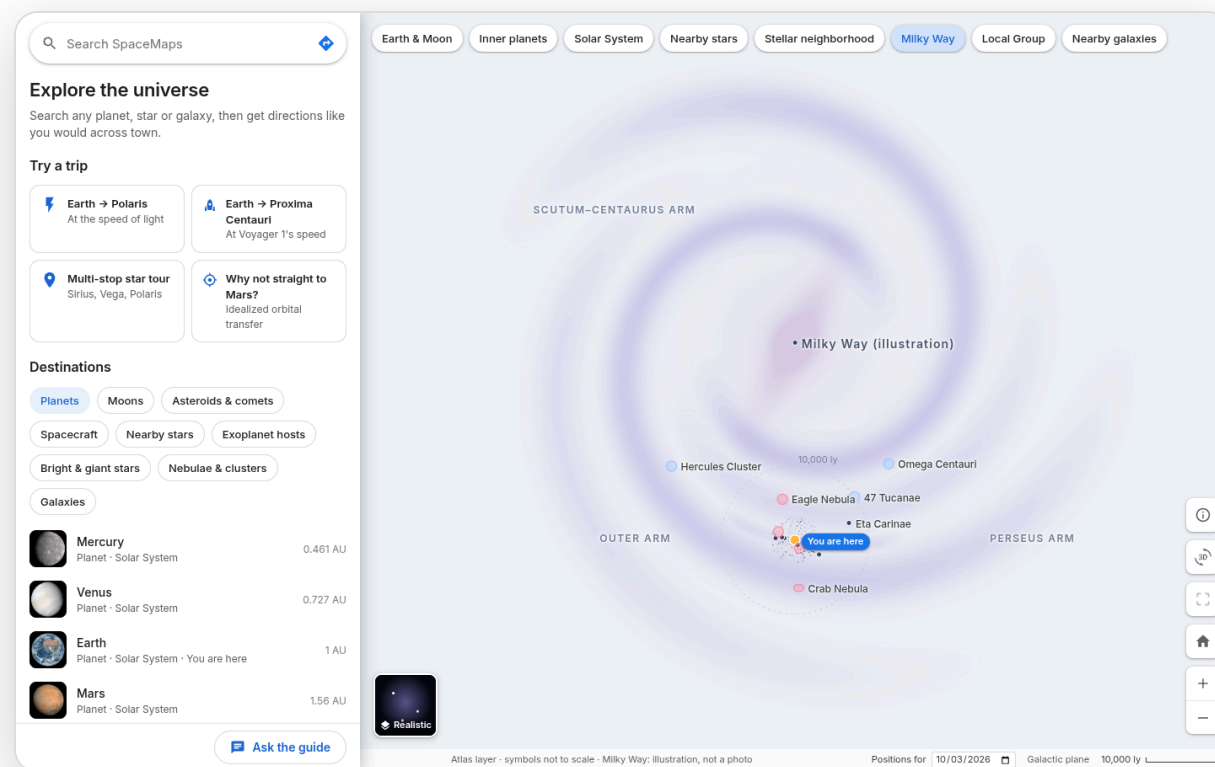
Why not fly straight to Mars?

- Idealized Hohmann transfer: **≈ 259 days**
- Mars must be about 44° ahead at launch, so windows open roughly every 26 months
- Circular, coplanar orbits; assumptions shown on screen
- An educational model, not a launch-ready mission

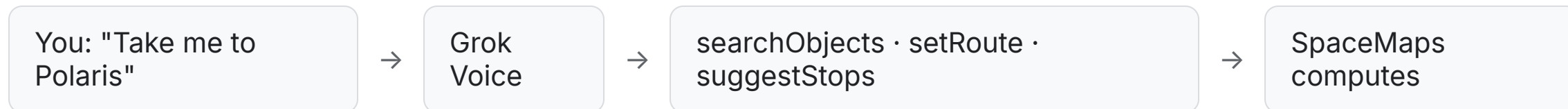


One map, 17 orders of magnitude

- Float64, camera-relative rendering (Three.js) from 6,000 km to 10^{21} km
- van Wijk–Nuij smooth zoom flights; ecliptic → galactic plane blend
- 109k stars in one batched draw, colored by temperature
- Reproducible data pipeline → 4.5 MB offline bundle
- 47 unit tests (physics, frames, dataset) + headless browser smoke test

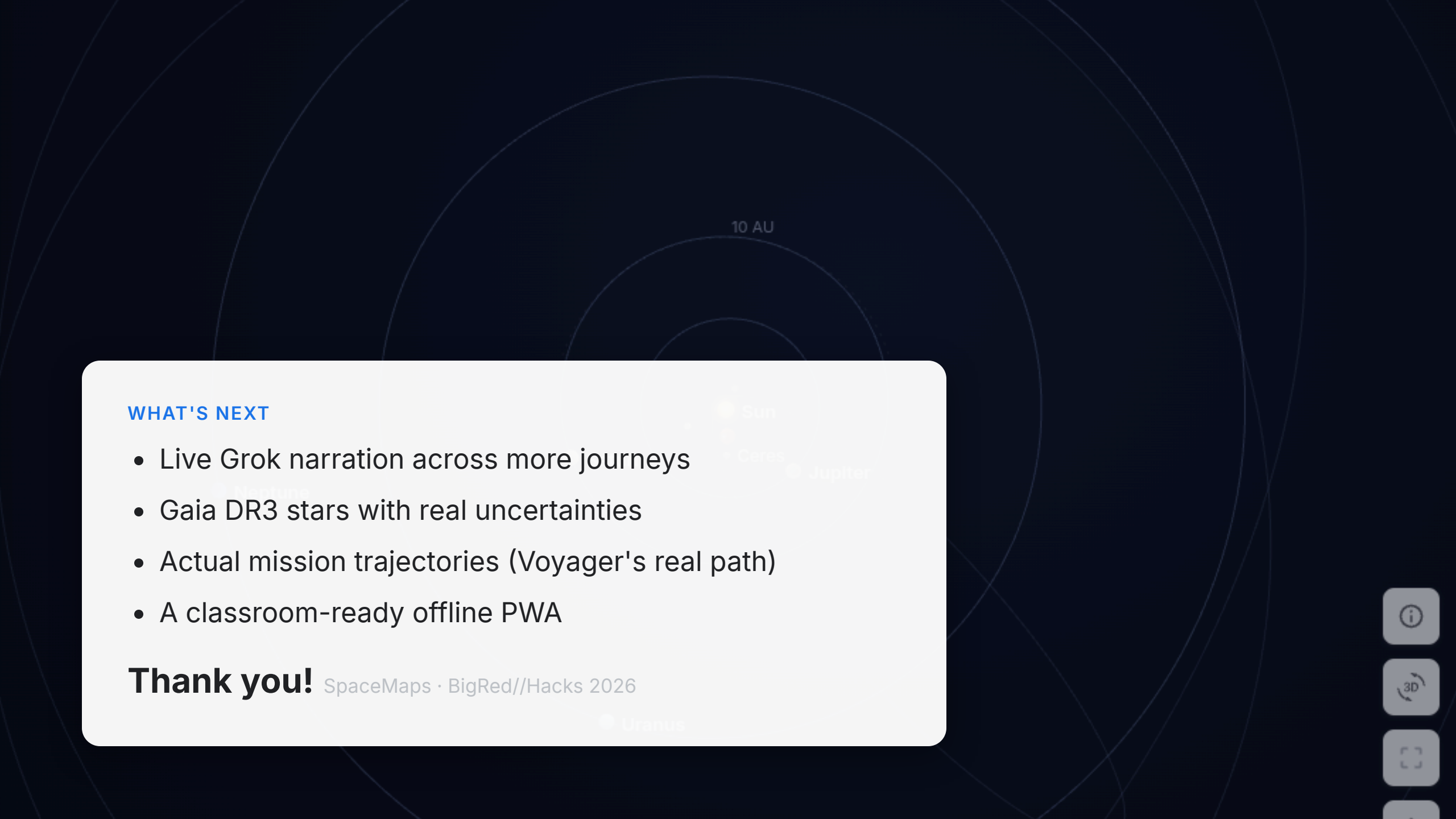


A grounded guide



- The model only calls validated tools; IDs and arguments are checked against the catalog
- The API key stays on the server; the browser gets a 5-minute ephemeral token
- Without a key: a clearly labeled *offline scripted guide*, never presented as Grok

Status: Grok Voice and Grok Imagine are implemented **not yet tested with a live API key**



WHAT'S NEXT

- Live Grok narration across more journeys
- Gaia DR3 stars with real uncertainties
- Actual mission trajectories (Voyager's real path)
- A classroom-ready offline PWA

Thank you! SpaceMaps · BigRed//Hacks 2026